

CALCULUS 101 · ANSWER KEY - PROFESSOR COPY

Homework 1: How Fast Is the Baby Growing?

Grading guide. Total 20 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Rates from a table

1. Three babies were weighed on day 1 and again on day 8. Fill in the change in grams, the number of days, and the rate in grams per day. (Change = later weight - earlier weight. Rate = change $\div$ days.) (6 pts)

Answer: A: change 70, 7 days, 10 g/day. B: change 140, 7 days, 20 g/day. C: change 0, 7 days, 0 g/day. 2 pts per baby (1 for the change, 1 for the rate).

Part B — Plot it

2. On the grid above, plot Baby A's weights as dots and connect them: day 1 — 800 g, day 3 — 820 g, day 5 — 840 g, day 7 — 860 g. Then answer: is the line steep or gentle? What is Baby A's rate in grams per day? (3 pts)

Answer: Four dots rising in a straight line (2 pts for plotting; 1 pt off per misplaced dot). Gentle/steady; 10 g/day (1 pt).

Part C — When the line goes down

3. A full-term newborn weighed 3,400 g at birth and 3,200 g on day 4. (a) What was the change? (b) What was the rate per day? (c) Should the team worry? (Hint: newborns normally lose up to about 10% of their birth weight in the first days, then regain it.) (3 pts)

Answer: (a) -200 g (a loss of 200 g). (b) -50 g/day (a loss of 50 g each day). (c) No — 200 out of 3,400 is about 6%, within the normal early loss; the team expects the line to turn back uphill within a week or two. 1 pt each.

4. On a weight graph, a steeper line means... (1 pt)

Answer: A. the weight is changing faster. - Steepness is the rate: how much the weight changes each day.

Part D — Per kilogram

5. A baby weighs 2 kg and gains 40 g in one day. What is that in grams per kilogram per day (g/kg/day)? Is it in the healthy 15-20 range? (2 pts)

Answer: $40 \div 2 = 20$ g/kg/day. Yes — right at the top of the healthy range.

Part E — Flashback

6. From NICU Tutor: what is the healthy growth target for a preemie, in g/kg/day? (1 pt)

Answer: About 15-20 g/kg/day.

7. From Biology 101: one cell goes through 3 rounds of division. How many cells now? (1 pt)

Answer: 8 (1 -> 2 -> 4 -> 8).

Part F — Think like a doctor

8. Two preemies both weigh exactly 1,500 g today. Baby X's weight line has climbed steadily all week. Baby Y's line has been flat for a week. Which baby worries you more, and what might the team do about it? (3 pts)

Answer: Baby Y (1 pt) — same weight today, but a flat line means no growth this week, while X is on track. The team would look for a reason (is the baby sick? using energy to breathe? not absorbing feeds?) and likely add calories — more milk, or fortifier added to the milk — then watch the line (2 pts for a reasonable plan mentioning a cause-check and/or more calories).